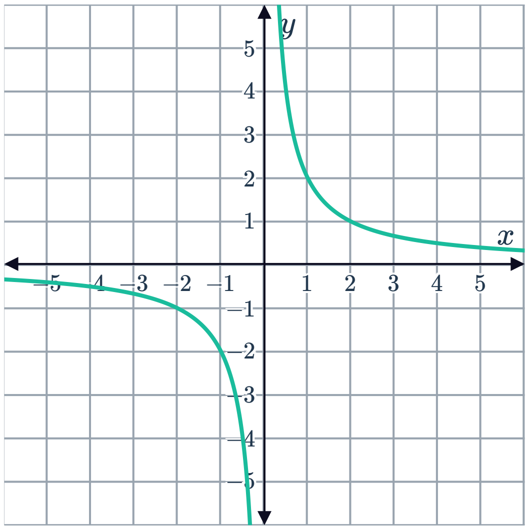
Key features of a hyperbola

1 Consider the graph of $y = \dfrac{2}{x}$.



- **a** For $x > 0$, as x increases, what does y approach?
- **b** For $x > 0$, as x approaches 0, what does y approach?
- **c** State the equation of the vertical asymptote.
- **d** State the equation of the horizontal asymptote.
- **e** The graph has two axes of symmetry. State their equations.

2 Consider the function $y = -\dfrac{12}{x}$.

- **a** When $x = 3$, what is the value of y?
- **b** If x is a positive value, is the corresponding y value positive or negative?
- **c** If x is a negative value, is the corresponding y value positive or negative?
- **d** In which quadrants does the graph of $y = \dfrac{-12}{x}$ lie?

3 Consider the function $y = -\dfrac{5}{x}$.

- **a** For what value of x is the function undefined?
- **b** As x approaches 0 from the positive side, what does y approach?
- **c** As x approaches 0 from the negative side, what does y approach?
- **d** As x approaches ∞, what does y approach?
- **e** As x approaches $-\infty$, what does y approach?

4 Consider the function $y = \dfrac{2}{x}$.

a Complete the table of values.

x	-2	-1	$-\dfrac{1}{2}$	$-\dfrac{1}{10}$	$-\dfrac{1}{100}$	$\dfrac{1}{100}$	$\dfrac{1}{10}$	$\dfrac{1}{2}$	1
y									

b For what value of x is the function undefined?

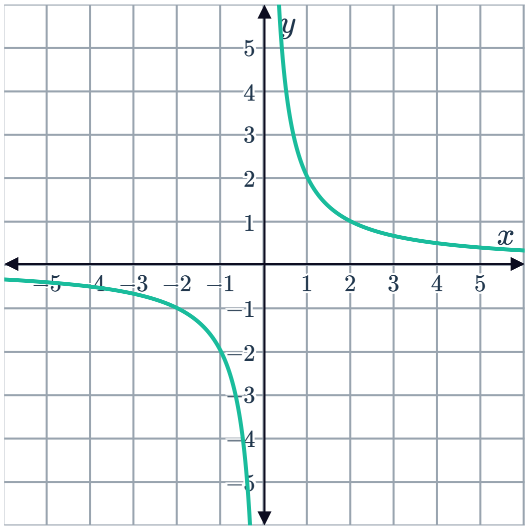
c Rewrite the equation to make x the subject.

d For what value of y is the function undefined?

e Below is the graph of $y = \dfrac{2}{x}$.
Complete the following statement:

As x approaches $\square$ from the right, the function value approaches ∞. As x approaches $\square$ from the left, the function value approaches $-\infty$.

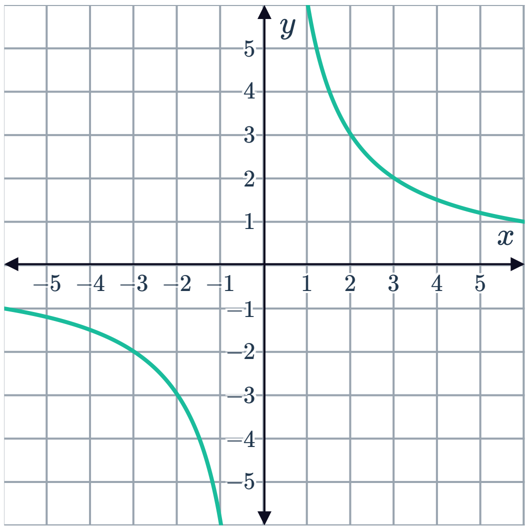
As x approaches ∞ and $-\infty$, y approaches $\square$. This is called the limiting value of the function.



5 Consider the hyperbola shown:

Complete the statement:

"Every point (x, y) on the hyperbola is such that $xy = \square$."



6 Determine whether the following relations[redacted] can be modelled by a function of the form $xy = a$?

a The relationship between the number of people working on a job and how long it will take to complete the job.

b The relationship between the number of sales and the amount of revenue.

c The relationship between height and weight.

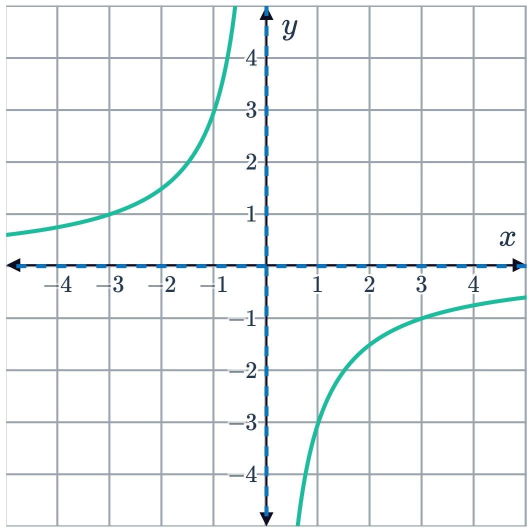
7 Consider the function graphed below:

a What is the equation of the vertical asymptote?

b What is the equation of the horizontal asymptote?

c State the domain of the function.

d State the range of the function.



8 For functions of the form $f(x) = \frac{k}{x}$, where k is a constant, the domain is always $(-\infty, 0) \cup (0, \infty)$. What does this mean for the function when $x = 0$?

9 Consider the function $y = \frac{1}{x}$.

a For what value of x is the function undefined?

b What is the domain of $y = \frac{1}{x}$?

c For what value of y is the function underfined?

d What is the range of $y = \frac{1}{x}$?

10 Patricia determined the range of the function $y = \frac{2}{x}$ is $(-\infty, 0) \cup (0, \infty)$.

By filling in the gaps below, complete her reasoning.

Looking at $y = \frac{2}{x}$, notice that the numerator ⬚ is non-zero, and so $\frac{2}{x}$ can never be equal to ⬚.

Another approach is to rearrange the equation and make x the subject. Then we get $x =$ ⬚. In this form we can see that the denominator ⬚ cannot be equal to ⬚.

11 Consider the functions $y = \frac{4}{x}$ and $y = \frac{2}{x}$.

a When $x = 2$, what is the value of y if $y = \frac{4}{x}$?

b When $x = 2$, what is the value of y if $y = \frac{2}{x}$?

c Which graph lies further away from the axes?

d For hyperbolas of the form $y = \frac{k}{x}$, as the k value increases, what happens to the graph?

Graphs of hyperbolas

12 Consider the function $y = \frac{6}{x}$.

a Complete the following tables of values:

x	$\frac{1}{4}$	$\frac{1}{2}$	1	2	4
y					

b Plot the points from the table of values on a cartesian plane.

13 Ursula wants to sketch the graph of $y = \frac{7}{x}$, but knows that it will look similar to many other hyperbolas.

What can she do to the graph to show that it is the hyperbola $y = \frac{7}{x}$, rather than any other hyperbola of the form $y = \frac{k}{x}$?

14 For each of the following functions:

i Complete a table of values of the form:

x	-2	-1	$-\frac{1}{2}$	$\frac{1}{2}$	1	2
y						

ii Sketch the graph of the function.

iii State which quadrants the function lies in.

a $y = \frac{2}{x}$

b $y = -\frac{1}{x}$

c $y = -\frac{1}{4x}$

Applications

15 The relationship between the current, C, (in amperes) and resistance, R, (in ohms) in an electrical circuit is given by:

$$C = \frac{240}{R}$$

where the voltage provided to the circuit is 240 V.

a Graph this function for $0 \leq R \leq 50$.

b What happens to the current as the resistance increases?

16 The rent, electricity, telephone bill and other expenses for a flat costs a total of $\$490$ per week. These expenses are shared equally between the tenants of the flat.

a How much will each occupant pay in dollars if the flat is shared by two people?

b Let the number of occupants be x, and the cost paid by each occupant be y.

Write a formula that relates the two pronumerals.

c Use this equation to fill in the following table of values.

x	1	2	3	4	5	6
y						

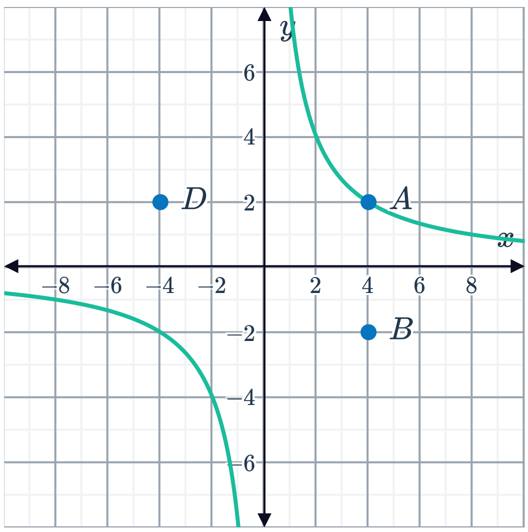
d Sketch the curve from your equation.

e What type of relationship exists between x and y?

17 Consider the hyperbola that has been graphed. Points $A\,(4, 2)$, B, $C\,(-4, k)$ and D form the vertices of a rectangle.

a Find the value of k.

b Hence find the area of the rectangle with vertices $ABCD$.



18 Boyle's law describes the relationship between the pressure and volume of a gas of fixed mass under constant temperature. The pressure for a particular gas can be found using

$$P = \frac{6000}{V}$$

where P has units $\mathrm{kg/cm^2}$ and V has units $\mathrm{cm^3}$.

a Graph the relationship $P = \dfrac{6000}{V}$ for $0 \leq V \leq 2000$.

b What is the pressure if the volume is $1\ \mathrm{cm^3}$?

c What happens to the pressure as the volume increases?

19 The time it takes a commuter to travel $100\ \mathrm{km}$ depends on how fast they are going. We can write this using the equation $t = \dfrac{100}{S}$ where S is the speed in $\mathrm{km/h}$ and t is the time taken in hours.

a Graph the relationship $t = \dfrac{100}{S}$.

b What is the time taken if the speed travelled is $10\ \mathrm{km/h}$?

c What is the time taken if the speed travelled is $50\ \mathrm{km/h}$?

d If we want the travel time to decrease, what must happen to the speed of travel?